Research and Design Document

Meal Planner App for the Elderly

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1. Introduction

1.1. Purpose of Research & Design Document

This document outlines the foundational research and high-level design for **Meal Planner App for the Elderly**. Its primary goal is to **record and justify the technical and strategic decisions** we have made before starting development. **After reviewing the literature, it became clear that simply listing features wouldn't be enough,** we needed to explain *why* a particular architecture or data strategy was chosen based on the project constraints.

While the Vision Document states *what* the app should do and the Specification details *how* it will work, **Research & Design** document **shows how the research on elderly nutrition and existing apps directly led to the proposed design.** It's the blueprint that ensures the final product is not just functional, but appropriately tailored to its users.

1.2. Scope of document

The scope for this project comes from a clear, practical problem which is observed during initial research**.**

Research behind the Decisions

- Investigate the specific nutrition challenges of elderly peoples.
- Evaluating technologies like **JAMSTACK** for their suitability for rapid, deployable prototype.

Resulting Design Blueprint

- The high level architecture shows how functionalities works combined.
- Planned data structure example, how a recipe/meal or user profile will be structured in code.

2. Domain research: elderly nutrition & meal planning

2.1. Nutrition challenges faced by elderly

Elderly peoples commonly face reduced appetite, changes in taste and smell, and earlier fullness, all of which lead to lower food intake. At the same time, ageing affects nutrient absorption due to changes in digestion and metabolism. Chronic illnesses and long-term medication use further complicate dietary needs and food choices. Physical limitations such as dental issues, swallowing difficulties, and reduced mobility can make eating and meal preparation more difficult. For the target users like **Paddy** described in vision, the social factor of living alone is a primary concern we must address, the app should reduce feelings of loneliness.[7][8][9]

2.2. Importance of nutrition rules and professional guidance

Good nutrition plays a critical role in maintaining health, independence, and quality of life in later years. Poor nutrition is strongly associated with muscle loss, reduced mobility, increased fall risk, and weakened immunity. Protein intake is particularly important, as older adults require relatively higher amounts to preserve muscle mass. Micronutrient deficiencies, especially vitamin D, calcium, and vitamin B12, are also common and often undiagnosed.

3. Review of existing applications

3.1. Overview of existing meal planner and health apps

Mealime is popular meal planning and recipe recommendation app designed to meal selection complications. It's core functionalities include curated recipe collections, auto generate shopping list based on meal selections and dietary preference. However, their approach is generic rather than personalized to specific nutrition or accessibility needs. [1][4][6]

3.2. Strengths and limitations of existing solutions

Strengths

- Visually clean interface with meal type and recipe cards.
- Integrated auto generate shopping list based on meal selections.
- Dietary customization e.g. vegetarian, gluten-free, dairy-free.
- Best option for general user but not specific to elderly.
- Option for serving size of meal.
- Allow user to change dietary preference anytime.

Limitations:

- **Bad Accessibility:**
  Small interactive elements e.g. checkbox, fonts size that make app difficult to use for elderly peoples.
- **Inefficient List:**
  Checking off items from shopping list is time taking and difficult with long lists, requiring manual checking one-by-one
- **No Barcode service:**
  Lack of barcode functionality to scan grocery items to add in list.
- **Meal Planning:**
  The user can just add meals into meal plane but without options for daily/weekly meal plan.

3.4. Lessons Applied To The Proposed System

- **Accessibility:**
  Large buttons and font, clear navigation to walk around app.
- **Structure Meal Planning:**
  Allow users to plan meals by daily/weekly plan.
- **Mult-User feature:**
  Support multi-user profile e.g. caretakers and family members.

4. Technology Research and Selection

4.1. Overview of Candidate Technology Stacks

The proposed application requires a technology stack that supports rapid iterations, simple deployment and perfectly aligns with agile methodology. The chosen technology must allow incremental features development and easy to update without large architectural challenges and also the frontend's accessibility for elderly users would need constant iterative refinement.

4.2. Justification for JAMstack Architecture

A JAMstack (javaScript APIs and Markup) based architecture was choose. The reason behind of chosen this stack is their simplicity, scalability and suitability for modern web applications. By separating frontend / UI form data and business logic, JAMstack enables a clean and modular system design.

4.3. Frontend Technology Selection

A modern javaScript based front end library React with Next.js framework was choose to support component based development and reusable user interface elements. It provides a file based routing system that simplifies the project structure and critically it has top level support for API routes, this mean we can build entire backend within the same project eliminating configuration hell. [1][2][3][4]

4.4. Backend and Data Handling approach

Following the JAMstack architecture, the backend logic will live as serverless functions (Next.js API Routes). This means we write Node.js functions for tasks like applying nutrition rules or filtering recipes, and Next.js / Vercel handle all the server infrastructure. This approach strongly supports the future integration with external nutrition and meal APIs while keeping initial implementation simple and in controlled.

4.5. Deployment Platform Selection (Vercel)

The deployment strategy is non-negotiable. For deployment Vercel is used. It is built by the creators of Next.js and offers seamless integration. The "git push" deployment perfectly aligns with the agile cycle every feature branch can have a live preview, and the main branch auto-deploys to a production URL. This guarantees that the development meet the core requirement of having a publicly accessible, deployed product at the end of every iteration that results in a potentially shippable product.

5. Data Source Strategy

5.1. Evaluation of External Meal and Nutrition APIs

During the planning phase of project, external meal and nutrition APIs were considered as a possible data source for recipes and nutritional information. While several well known APIs exist, but most of them are for commercial use with strict rate / API call limits or require paid subscriptions for use.

According to the nature of the project and the limited timeframe depend on external APIs from the beginning was considered a high risk issues such as cost, availability and dependency on third party services can negatively impact development and testing phase.

5.3. Use of Mock Data in Early Iterations

To meet the need of controlled, reliable and cost free source for data of data, the iteration 1 will use mock data which is store locally as structured JSON file. This approach ensures meal planning logic can be developed and tested without external API dependencies, costs, or network issues

5.4. Strategy for Future API Integration

The initial version of the system depend on locally managed data, the application designed with future API integration. Data access for the system will be keep separate from business logic, So, it will be easy to integrate API in future iterations. The system will be designed in a way that it will still apply same nutrition rules filtering logic while integrating API. This approach allows the project to remain flexible and scalable in future updates.

6. High-Level System Architecture

6.1Overview of System Architecture

The system follows a decoupled, 3-tier architecture aligned with JAMstack architecture which cleanly separates the user interface, business logic, and data management into different layers that communicate with each other through well defined interface primarily API calls.

The primary goal of this architecture are:

Maintainability:

To allow the frontend UI and backend logic to be developed , tested, and update independently.

Deployable Product:

To enable straightforward, automated deployment of a static frontend and serverless backend fulfils the **Potentially shippable product** requirement at every iteration.

Extensibility:

To make any changes in any layer e.g. swapping mock data with external API with a minimal change.

6.2. JAMstack Architectural Overview

The chosen architecture is the modern classification of JAMstack implemented using Next.js framework. The architecture is visualize in the diagram

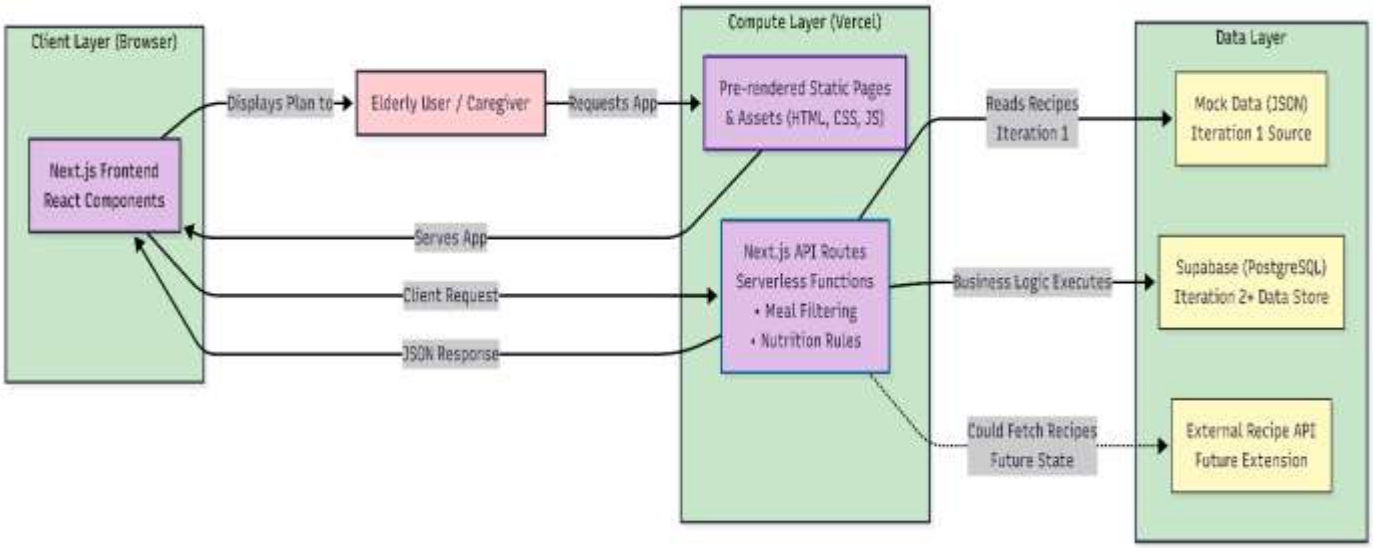

Fig.1. System Architecture Diagram [Author's own]

Key Data Flow:

- **Request:** User accesses the public URL. Vercel's Content Delivery Network CDN serves the pre-built, static HTML / CSS / JS.
- **Interaction:** The React frontend (hydrated in the browser) captures user input (e.g., profile setup).
- **Processing:** The frontend calls a specific Next.js API Route (e.g., POST /api/generate-plan), sending user data.
- **Execution:** The serverless function executes the business logic, it fetches recipe data from the current source (mock JSON or Supabase), applies the filtering algorithm, and generates a plan.
- **Response:** The function returns the meal plan as JSON to the frontend, which renders it for the user.

6.3. Separation of Concerns (UI, Logic, Data)

The chosen architecture enforces a clean separation of concerns, which is critical aspect of agile development environment.

Layer	Technology	Responsibility
Presentation UI	React / Next.js	Display all the user interfaces and data Manage states, client side validation
Business Logic	Next.js API Routes	Contain core system rules Applies nutrition rules and filters Execute meal selection algorithm
Data Access & Storage	JSON Files / Supabase	Provides consistent interface to data, whether from local files or database

Table.1. Architecture Layer [Author's own]

The separation means that:

- The UI can refine Without Understanding the business logic.
- The data source can be change by updating one module who serves data without UI or business logic.

6.4. External Services and Future Extensibility

The current version of system does not uses external services for its core functionality like data source. All meals and nutrition data is managed locally during the early development phase and then later stored within the system's database. Thius ensure the system still remains stable and independent of third party source availability.

However, the system is designed to adapt and support future extensions such as external services e.g. recipes or nutrition APIs, can be integrated in the system without making major changes in overall system architecture.

7. Major System Components and Interfaces

7.1. User Interface Layer

The user interface layer is responsible for displaying information to the user and collecting user input via forms. It is implemented using reusable components based framework and simple layouts suitable for elderly users and caregivers.

Primary Responsibilities:

- Rendering all application screens and navigations.
- Capture user's input through forms.
- Displaying outputs such as meal plans and shopping list.

7.2. Business Logic Layer

The business logic layer contains the core logic of the application acts as system's decision-making engine. It combines all nutrition rules, filtering meals based on user preferences, and meal generation logic.

Primary Responsibilities:

- Processing user profiles to apply nutrition rules.
- Coordinating with data access layer for data needs.
- Execute meal generation and shopping list logic.

7.3. Data Access Layer

The data access layer is responsible for retrieving and storing application data. In early development stages, this data is read from locally managed JSON files. In later iterations, the same data is stored in a database to support persistence and more realistic system behaviour. The data access layer ensures that changes to the data source do not impact other parts of the system.

Primary Responsibilities:

- Retrieving meal data from the current data source (mock dataset or live database)
- Carry and retrieving user's profile information
- Manage data for shopping lists and meal plans.

7.4. Interfaces Between Components

Communication between system components is handled through clearly defined interfaces. The interaction between layers follows a clear, unidirectional flow to maintain architecture integrity.

Key Interface Contracts:

1. **UI → Business Logic Interaction:** The UI layer communicates with the business logic via structured API calls.

2. **Business Logic → Data Access Contract:** The business logic interacts with the data layer through a defined data service interface. This interface specifies methods for querying recipes and managing user data without exposing storage details.
3. **Shared Data Models:** Consistent data structures form the shared vocabulary across all layers. These models are defined conceptually in Section 8 and ensure data consistency across boundaries.

8. Data Design and Major Data Structures

8.1. Conceptual Data Model Overview

The data model for the Meal Planner App is designed around four core entities, Meal, User Profile, Meal Plan, and Nutrition Rule.

Relationship between entities from prospect of system logic:

- A user profile defines the preference.
- The system evaluates available meals against these preferences.
- Approved meal by nutrition rules are assembled into meal plan.

8.2. Meal Data Structure

Attribute	Description & Purpose
Identifier	Uniquely identifies each meal within the system.
Name	Meal name shown to users in the interface.
Meal Type	Indicates when the meal is intended to be consumed (e.g. breakfast, lunch, dinner).
Description	A short text explaination of the meal to help users understand the option.
Ingredients	A list of ingredients associated with the meal, used for planning and shopping support.
NutritionalInfo	A reference to the meal's nutritional data used for rule evaluation.
Allergens	Information about allergens present to ensure user safety.
Dietary Tags	Labels used to support preference-based and medical filtering.
Cultural / Cuisine Category	Indicates the cuisine or cultural style of the meal.

Table.2. Meal Data Structure [Author's own]

8.3. Nutritional Information Structure

NutritionalInfo object is sub-structure of meal stored separately from meal descriptions to keep concerns clearly separated and to enable consistent rule evaluation across different data source.

Table 8.3 summarises the core attributes of the nutritional information structure and their purpose within the system.

Attribute	Description & System Purpose
Nutritional ID	A reference used to uniquely identify a set of nutritional data.
Carbohydrates	Indicates carbohydrate content for balanced meal evaluation.
Protein	Represents protein content, which is a key focus for elderly nutrition rules.
Fat	Indicates fat content used for general dietary balance checks.
Sodium	A critical micronutrient used for condition-based filtering.
Calcium	A key micronutrient relevant to bone health in elderly users.
Nutrient Source Reference	Indicates the origin of the nutritional data (e.g. guideline-informed, curated dataset).

Table.3. Nutrition Information Data Structure [Author's own]

8.4. User Profile and Preferences

The user profile stores information required to personalise meal recommendations and apply nutrition rules appropriately.

Table 8.4 outlines the key attributes included in the user profile and their role within the system.

Attribute	Description & System Purpose
User Identifier	A unique reference used to distinguish each user profile.
Age Group	Indicates the user's age range, used for applying age-related nutrition considerations.

Attribute	Description & System Purpose
Dietary Preferences	User-declared food preferences, such as vegetarian or cultural choices.
Allergies	Safety-critical information used to exclude unsuitable meals.
Health Considerations	Relevant health conditions that influence nutrition rule application.
Budget Preference	A general indicator used to support cost-aware meal planning.
Meal Preferences	User likes or dislikes used to refine meal selection.

Table.4. User Profile Data Structure [Author's own]

8.5. Weekly Meal Plan Structure

A weekly meal plan represents the outcome of the meal recommendation process. It groups selected meals by day and meal type over a one week period.

Table 8.5 summarises the main elements of the weekly meal plan structure and their purpose within the system.

Attribute	Description & System Purpose
Plan Identifier	A unique reference used to identify a generated weekly meal plan.
User Reference	Links the meal plan to a specific user profile.
Daily Entries	Groups meals by day of the week.
Meal Slots	Organises meals within each day such as breakfast, lunch, dinner.
Meal References	Links to the selected meals included in the plan.
Nutrition Summary	A high-level overview of nutritional coverage across the week.
Generation Timestamp	Records when the plan was created or last updated.

Table.5. Weekly Meal Plan Data Structure [Author's own]

9. Key Algorithms and Decision Logic

9.1. Overview of Meal Recommendation Logic

The meal recommendation process is designed to select suitable meals based on user's preference and predefined nutrition rules. The system evaluates available meals against user preferences and constraints before identifying options that are suitable for meal plan. The goal of this process is to balance nutritional suitability with personal preferences.

9.2. Nutrition Rule Evaluation Approach

Nutrition rules are applied as part of the decision making process rather than being hardcoded into the data itself. These rules are derived from professional guidelines and are used to assess whether meals meet basic nutritional requirements or violate health-related constraints.

9.3. Meal Filtering and Selection Process

Meal filtering is performed by excluding options that conflict with user-defined constraints, such as allergies or dietary restrictions. Remaining meals are then evaluated for suitability based on nutrition rules and user preferences. This step reduces the available set of meals to those most appropriate for the individual user.

9.4. Weekly Meal Plan Generation Logic

The weekly meal plan is generated by organising selected meals into daily meal slots over a fixed period of time. The system ensures that each day contains a balanced set of meals while maintaining consistency with user preferences and nutritional considerations. If user information or rules change, the plan can be regenerated using the same logic.

10. Design Constraints and Assumptions

The Design Constraints and Assumptions section outlines the foundation limits and accepted conditions that shapes the system implementation and design.

10.1 Accessibility and Usability Constraints:

The system is designed by follows accessibility and usability standards.

- **Visual Accessibility:**
  Visual Focus:
  All interactive elements should have highly visible focus indicator. All the user interface must follow the WCAG (Web Content Accessibility Guidelines) 2.1 level AA guidelines.
  Contrast Ratio:
  Text must have a minimum contrast ratio of 4.5:1 against background.
  Text Size: Use base font size of at least 16pixels functionality loss.

- **Minimizing Physical and Mental Effort:**
  Touch targets:
  All buttons and interactive elements should have minimum touch target size of 44x44 pixels.
  Error Tolerance:
  The system must prevent errors where possible and guide users with simple and plain language instructions.

10.2 Technical Constraints

The technical implementation of the system is strictly followed by chosen architecture, platform and project scope to ensure delivery of project with given timeline.

- **Deployment & Delivery:**
  The application must be deployed as a publicly accessible, mobile-first web application after each iteration. This mandates the use of a platform like Vercel for zero-configuration deployment.
- **Architecture:**
  The system must adhere to the JAMstack architecture using Next.js. This constraints backend logic to serverless functions API Routes.
- **Development Scope:**
  The prototype will not implement user authentication, payment systems, or real-time collaborative features e.g. live chat between user and caregiver. These are out of scope for the academic prototype

10.3 Data And Dependency Constraints

Data handling is constrained by practical considerations of cost, reliability, and project risk management.

- **Initial Data Source:**
  For Iteration 1, the system cannot depend on any external, live APIs for core recipe data. All data must be sourced from locally managed mock datasets (JSON files) to guarantee uptime, avoid costs, and enable deterministic testing.
- **Nutritional Accuracy Liability**:
  The system must include a clear disclaimer that its meal recommendations are based on prototype data and simplified rules. It is not a certified medical device and does not replace personalised advice from a qualified nutritionist or doctor.
- **Browser Compatibility:**
  The frontend must be fully functional on the latest versions of Chrome, Safari, and Firefox for both desktop and mobile, reflecting the most likely usage environments for the elderly demographic.

10.4 Key Assumptions

The design and planning are based on the following accepted assumptions:

- **User Environment:**
  The target user has access to a stable internet connection and a smartphone or tablet with a modern web browser.

- **Data Input:**
  It is assumed that the user (or their caregiver) can accurately input their dietary restrictions and preferences into the application.

- **Iterative Feedback:**
  The design assumes that usability feedback will be gathered after each iteration from supervisors and, if possible, target user proxies, and that this feedback will inform subsequent design refinements.

11. Conclusion

The rationale and planning for the Meal Planner App for the Elderly have been described in this Research and Design document. A assessment of current meal planning apps and domain research on senior nutrition revealed important gaps in older user suitability, customisation, and accessibility. The design objectives and system architecture suggested in this project were directly influenced by these findings.

A JAMstack-based solution that facilitates simplicity, scalability, and iterative development was chosen after pertinent technologies and architectural approaches were assessed. System architecture, component separation, data strategy, and core algorithms are examples of high-level design choices that were documented to guarantee that development is directed by well-defined and supported design concepts rather than ad hoc implementation.

Together with Vision and Specification documents, R&D documents provides a structured foundation for building a function, accessible and maintainable system.

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DECLARATION

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*I understand that I can only get credit for work that I have done (e.g. understanding other code, interfacing my code with it, modifying it, etc.), but you will not be credited with the third-party code itself.

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